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Sweet escape: The role of empathy in social media engagement with human versus virtual influencers

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ABSTRACT

Virtual influencers engage in emotional sharing to gain and keep followers. However, given that many people use social media for diversion purposes, this emotional sharing may hinder users' ability to escape from everyday emotional experiences, particularly for highly empathetic individuals. Using a between subjects, randomised experimental design, we explore how empathy affects reactions to virtual vs. human influencers, showing that those highest on empathy are more likely to follow a virtual influencer, and rate her as more socially attractive, than a comparable human influencer; these results disappear when the influencers' true nature is unknown to participants. We postulate that these results represent an "escapism effect", where the virtual influencer is expected to provide greater diversionary benefits from everyday human emotional experiences and require fewer cognitive resources in the form of emotional sense making. We present practical implications and future research opportunities arising from this effect.

1. Introduction

The social sharing of emotions is a common human tendency, one that predates the use of social media and mobile devices (Rimé, 2009). The democratisation of the Internet has enabled individuals to move this social sharing online and the evolution to Web 2.0 has given rise to platforms such as Twitter and Instagram, which showcase individuals' emotional experiences on a much larger scale (X. Liu et al., 2020). Although digital entities such as chatbots and avatars are already prevalent on social media (Nowak et al., 2015; Xu et al., 2017), we have recently seen the emergence of virtual agents in these human networks, who operate independently from user input (Arsenyan & Mirowska, 2021; Park et al., 2021). In the form of virtual influencers, virtual agents are already used for marketing purposes (Tan & Liew, 2020; Yen & Chiang, 2021) and are quickly becoming opinion leaders in areas of fashion and entertainment (Tietjen, 2018). These virtual agents are created to blend into the social network in which they operate, in order to compete with their human counterparts in a context of short attention spans and abundant content (Chang, 2010) and, to this end, may seek to mimic the human sharing of emotional experiences (Arsenyan & Mirowska, 2021).

However, the rise in popularity of social media influencers has not been without criticism. Detrimental effects on followers in terms of body image issues, excessive social comparisons, and depression (Alfasi, 2019) have resulted in questions being raised regarding the appropriateness of influencer use, particularly in marketing campaigns, given that social media use continues to grow amongst youth segments of the population. As virtual influencers are increasingly deployed on mainstream social media platforms, they may present a viable alternative to human influencers. Therefore, it is important to understand what reactions to virtual vs. human influencers are to be expected (Arsenyan & Mirowska, 2021) and how they may be affected by individual differences (von der Pütten et al., 2010). Identifying individual differences in willingness to engage with virtual influencers may allow for virtual influencers to be designed and deployed in a way that protects potentially vulnerable individuals (Vossen & Valkenburg, 2016).

Individuals engage in social media usage for specific purposes (McLaughlin et al., 2022). One of the primary reasons reported for social media use is diversion and the ability to experience emotional escape from one's regular life, trials, and tribulations (Huang & Su, 2018; Schmuck, 2021; Sheldon & Bryant, 2016). However, individuals' pursuit of diversion may be hindered by influencers' attempted emotional sharing as it affects their ability to emotionally disconnect when interacting with other users. This ability to emotionally disconnect may, in

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turn, be affected by individual level empathy – an individual's ability to "step into a character's life" (MacDorman, 2019, p. 140) and experience, understand, and/or distance themselves from the other's emotional experience (Carré et al., 2013, p. 20).

We posit that virtual influencers may provide a potential "sweet escape" for those highly susceptible to being drawn into others' emotional experiences by providing entertainment content that does not elicit the same level of emotional entanglement and sense-making as that provided by human influencers. We use a between subjects, randomised experimental design, to examine differences in willingness to engage with a virtual vs. human influencer, both when participants were aware of the influencers' true nature and when they were not. We look for interaction effects between experimental conditions and participant empathy scores to explore whether individuals who are more susceptible to being caught up in others' emotional experiences are more likely to follow a virtual influencer, and rate her as more attractive, than a comparable human influencer.

In the sections that follow, we will define the phenomena of virtual influencers and present our hypotheses and experimental methodology. We conclude with a discussion of the results, implications, and potential future research directions for this burgeoning field of study.

2. (Virtual) influencers

Social media influencers are micro-celebrities who specialise in cultivating a unique public image transmitted via their online presence on one or more social media platforms (Djafarova & Bowes, 2021; Khamis et al., 2016; Schmuck, 2021). Influencers succeed in creating a personal brand (Abidin, 2018), thereby transforming themselves "from citizens to corporations" (Senft, 2013, p. 351). This success, in turn, provides opportunities for income-generation as companies and brands seek to reach target audiences via the influencers' social media reach. As such, the activity of influencers has global economic, legal, social, and cultural consequences (Abidin, 2018).

More recently, the social media world has seen the rise of virtual influencers (VIs), an amalgamation of digital avatars and virtual agents (Arsenyan & Mirowska, 2021), who present a crafted reality not dissimilar from the manufactured authenticity of their human counterparts, but with the key distinction of being *authentically fake* (Wills, 2019). VIs are presented as human-adjacent, engaging in activities such as modelling, singing, socialising, and opinion-sharing (Arsenyan & Mirowska, 2021). Similarly to human influencers (HIs), VIs create their own public personas and elaborate storylines presenting varying degrees of anthropomorphism (Hanus & Fox, 2015), offering a potential alternative for both companies (seeking product placement opportunities) and users (as influencers to follow within various domains).

Since the popularisation of the first widely known VI, Lil Miquela (Shieber, 2018), several additional VIs have been crafted and launched in the digital arena (Ahn et al., 2022). From Noonouri, a computer-generated character stylized with anime features, to Shudu, the world's first digital supermodel, they represent the extremes of anthropomorphizing of virtual agents. Similar to their human counterparts, these VIs engage in the practice of microcelebrity (Marwick, 2018). As the influencer strives to appeal to audiences (by feeding content to) and communities (engaging in content with), the follower becomes an active agent, and even the commodity, in the exchange (Senft, 2013). By deciding where their much-valued attention (commodified in the form of clicks, likes, and comments) is directed, followers dictate which influencers and narratives get more attention and which are ignored. As such, VIs in highly visible human networks create a whole new context in which the Uncanny Valley Hypothesis can be applied (UVH; Mori et al., 2012). UVH argues that, as the human-likeness of an artificial entity increases, peoples' responses shift from affinity to eeriness as the entity, despite showing a lifelike appearance, is not quite human. Even though Mori (1970) makes a distinction between mobility (e.g. robots) and stillness (e.g. images) in eliciting eeriness, the UVH has more recently been applied to computer generated and moving images (Seymour et al., 2017).

Regardless of their level of human resemblance, VIs attempt to recreate a similar experience for users as HIs, with emotionally charged content, realistic-sounding story lines, and even producing music and video content (Ahn et al., 2022; Arsenyan & Mirowska, 2021; Thomas & Fowler, 2021). Yet, it is not clear whether these attempts on the part of VIs to use similar means as humans to make an emotional connection with their audience have uniform effects for different types of users.

Previous research looking at the reactions of users to VIs on Instagram has relied on users who had already chosen to engage with the VI in some form. Arsenyan and Mirowska (2021) found that, although VIs attempted to mimic the positive emotions that are prevalent on Instagram, they earned more derogatory and sceptical language in the comments from the audience compared to their human, as well as more cartoon-like, counterparts. Additionally, despite earning higher traditional engagement metrics, such as likes and views, the human-adjacent VI garnered fewer positive emotional reactions than the human-like and cartoon-like influencer, even showing an "uncanny hill" with respect to higher negative reactions as compared to the other influencers in the study (Arsenyan & Mirowska, 2021). Other research has similarly found that VIs can elicit scepticism on the part of the audience that knows their true nature (Sands et al., 2022), and even affect attitudes toward the brand they are endorsing (Thomas & Fowler, 2021).

In contrast, we are interested in first impressions of HIs vs. VIs – that is, reactions on the part of people who do not yet follow an influencer and are not aware of their existence. Evidence suggests that such first impressions are key to the formation of behavioural and affective reactions (Lindgaard et al., 2006), which can in turn shape the evolution of the relationship between user and influencer, such as the likelihood and frequency of subsequent encounters (Cafaro et al., 2016). Affective reactions can include perceived compatibility with one's self and social circle, also known as social attractiveness (Spence et al., 2019), while behavioural reactions, such as willingness to affiliate (Philipp-Muller et al., 2020), can affect an influencers' engagement metrics (Taillon et al., 2020), and therefore their earnings potential (Abidin, 2018). Although limited research exists to date looking at VIs, evidence suggests that users have very different reactions to HIs vs. VIs, particularly when the VI has a human-like appearance (Arsenyan & Mirowska, 2021) or when a transgression, such as a politically incorrect statement, has been committed (Thomas & Fowler, 2021).

3. Hypothesis development

3.1. Need for diversion and emotional contagion

Uses and gratification theory posits that people actively calibrate their social media activity based on specific needs, such as the search for diversion from their offline lives (Sheldon & Bryant, 2016; Yen & Chiang, 2021). To this end, social media, particularly Instagram, are reported as important platforms for those seeking escape from their everyday problems and experiences (Huang & Su, 2018). However, exposure to emotionally charged content may actually hinder such escape, particularly for people who are highly susceptible to being caught up in the emotions on display (Weiss & Cohen, 2019). Nevertheless, HIs often play on emotional bonds with users to gain followers and build parasocial relationships to increase their visibility and online status (Hwang & Zhang, 2018), and VIs similarly attempt to use emotion to build bonds with followers (Arsenyan & Mirowska, 2021).

However, the UVH effect, where VIs who appear too human-like are dehumanised and referred to as "robots" or "fake", is pervasive on social media (Arsenyan & Mirowska, 2021). The use of such dehumanising language may point to expectations that any emotionally charged content posted by a VI will not have the same effect as that posted by fellow humans (Nowak et al., 2015). Therefore, we explore the role of empathy, and how it may interact with knowledge about an influencer's human

vs. virtual origins, in predicting individuals' willingness to follow and attractiveness of HIs versus VIs.

The popularly used term empathy is seen as a "fundamental skill for navigating everyday life" (Cameron et al., 2019, p. 4) and its absence has been linked to problematic behaviours and depressive states (Carré et al., 2013). Empathy serves as an important foundation in building a variety of relationships (Bickmore & Picard, 2005) and it is considered a key factor in audience reactions to human versus virtual agents (Misselhorn, 2009). Interestingly, Worral et al. (2021) found a lack of empathy when studying user motivations to engage in online interactions, although in follow-up questioning some of these users highlighted the empathetic dimension of their participation. These results suggest a subconscious desire to "shut off" ones' empathetic tendencies in certain contexts, something that may be much more easily justifiable when faced with a non-human agent's emotional experiences

As a higher order construct which encompasses multiple dimensions (J. A. Hall & Schwartz, 2019), empathy has traditionally been understood to include an affective and cognitive dimension (D'Ambrosio et al., 2009; Wieseke et al., 2012). Affective empathy, or emotional contagion, is defined as the "transfer of emotional states from one person to another" (Czarna et al., 2015, p. 318; Verhagen et al., 2014). It is considered an integral part of empathy, having been included in both two and three-dimensional models of the construct (Carré et al., 2013; D'Ambrosio et al., 2009; Jolliffe & Farrington, 2006). It is considered an automatic, bottom-up process, where individuals reproduce the emotions that another is experiencing within themselves (Carré et al., 2013). Of course, this can refer to the experiencing of both positive and negative emotions.

For individuals high on emotional contagion, who easily experience others' affective states, this empathetic reaction may block their ability to gain the true diversion benefits of social media use. The above-mentioned dehumanisation of VIs may serve to dampen one's empathetic response, as the terms found to be often used to refer to VIs by Arsenyan and Mirowska (2021) serve to place the VI in a category separate from the human user. This distancing may work in favour of users high on emotional contagion, in that they will expect to gain a greater escape experience from following a VI than a HI:

Hypothesis 1. *(H1):* Individual level emotional contagion will moderate the effects of influencer type on a) willingness to follow and b) attractiveness of the influencer, such that those higher on emotional contagion will be more willing to follow the virtual influencer and find her more attractive than the human influencer.

3.2. Resource conservation and cognitive empathy

The second dimension of empathy, cognitive empathy, is the ability to understand and reason about the affective states of others. Considered less automatic than contagion, this dimension of empathy activates brain regions responsible for the processing of environmental stimuli (Carré et al., 2013). Importantly, cognitive empathy should specifically involve an understanding of emotional situations, not just a broader recreation of the emotional experience or adoption of others' positions (Bensalah et al., 2016).

Recent research has found that empathetic reactions are cognitively taxing (Cameron et al., 2019) and it is well established that individuals prefer to conserve resources when their expenditure is not expected to lead to resource gains (Hobfoll, 1989). As argued above, the dehumanisation of VIs may serve to lighten the cognitive load required of users when interacting with them, as their experience is seen as less real and therefore not requiring in-depth processing or understanding. Therefore, in order to meet their goals of diversion, users high on cognitive empathy may expect to experience a greater emotional escape as a result of interacting with VIs rather than HIs:

Hypothesis 2. *(H2):* Individual level cognitive empathy will moderate the effects of influencer type on a) willingness to follow and b) attractiveness of the influencer, such that those higher on cognitive empathy will be more willing to follow the virtual influencer and find her more attractive than the human influencer.

3.3. Escapism and emotional dissociation

More recently, scholars have expanded to a three dimensional model of empathy, which has been validated with psychometric and neuropsychological evidence (Carré et al., 2013). The third dimension, emotional disassociation, is defined as the ability to disengage and protect oneself from others' affective experiences. As a top-down process, disassociation is a self-regulatory process drawing on areas of the brain responsible for executive function and the attending elements of self-control (Carré et al., 2013; Neumann et al., 2015; Yuan & Raz, 2014). For those low on this dimension, exposure to others' emotional experiences may put them at risk for bearing the brunt of the emotional experience. Therefore, such individuals may pursue "proactive person-environment interactions" (Mehl et al., 2006, p. 872), preferring to put themselves into situations suited towards their personal characteristics (Wilt & Revelle, 2019). In the case of a low ability to disassociate oneself from others' emotional experiences, the VI, given its non-human nature, may be a more attractive target of engagement if their social media presence is expected to be either lower in emotional content or less invasive as an emotional experience for the observer:

Hypothesis 3. *(H3)*: Individual level emotional disassociation will moderate the effects of influencer type on a) willingness to follow and b) attractiveness of the influencer, such that those lower on emotional disassociation will be more willing to follow the virtual influencer and find her more attractive than the human influencer.

3.4. Awareness

The above arguments depend on the social media user being able to discern an influencer's true nature. If an individual is seeking out profiles that will allow them some degree of escape from the emotional entanglement that their high empathy provokes in interactions with human profiles, then this escape is most likely to occur when the user is sure that the profile is not a human. We therefore propose one final hypothesis:

Hypothesis 4. *(H4):* Hypotheses 1, 2, and 3 will only be supported when participants are explicitly aware of an influencer's human versus virtual nature.

4. Methodology

We tested our hypotheses using a between-subjects, random assignment experimental design. The main variable manipulated was exposure to visual content of an HI versus VI. We conducted the experiment under two conditions: one where participants were made aware of the human vs. virtual nature of the influencer they viewed, and another where this information was not shared with participants.

4.1. Participants

Participants included 224 young adults who were given the opportunity to participate in the study in exchange for points towards course credit on a university-wide research participant recruitment platform. Ten participants reported enough familiarity with the social media influencers to whom they were exposed that their responses were removed from the study. The final sample consisted of 214 participants, majoring in Marketing, 37% female, and on average 22 years old (s.d. = 1.67), which is in the typical age range for Instagram users (Statista, 2019). All participants reported using some form of social media, with the majority accessing their social media accounts daily (21%) or several times a day (76%). Almost 80% of the participants reported spending at

least one hour per day on social media, with over a quarter reporting spending over three hours per day. Additionally, 86% of participants reported using the target social media of this experiment, Instagram, regularly. Please see Tables 1 and 2 for additional descriptive information about the final sample.

4.2. Procedure

4.2.1. Choice of virtual and human influencers

For our experimental stimuli, we chose pictures posted on Instagram for several reasons. Instagram is a highly visual platform (Casaló et al., 2020; Sheldon & Bryant, 2016; Waterloo et al., 2018), which includes more personal and intimate content based on one's personal (rather than relational) identity (Sheldon & Bryant, 2016). It is also the fastest growing and most used social media (Sheldon & Bryant, 2016) and particularly popular with influencers (Johnson et al., 2019). Finally, most users choose to maintain their information as publicly visible on Instagram (Waterloo et al., 2018), making any profile accessible to anyone.

Prior to launching the focal study, we asked a sample of 402 Instagram users recruited from the Prolific online research participant pool (45.5% female, average age 23.31, $SD = 4.00$), to evaluate six potential VI profiles for human likeness, eeriness, and likeability (Katsyri et al., 2017). All six of the VI profiles chosen depicted influencers presenting as female. This was due to the fact that the majority of Instagram users and influencers are female (Djafarova & Rushworth, 2017; Roberti, 2022). Additionally, at the time of this study, there were few male virtual influencers active on social media, which would have unduly limited the pool from which we could select an appropriate VI.

In line with the UVH (Mori, 1970), it was decided that the VI scoring the lowest on these dimensions may be too easily confounded with a human, while the VI scoring the highest on these dimensions was too obviously a virtual entity to allow for valid comparisons with an HI. We therefore chose Lil' Miquela (LM; Fig. 1) as the focal VI given that her scores on all three dimensions were closest to the average across all of the VIs.

Our next step was to identify an HI with a similar social media presence: single, female, 18 to 25 years old with artistic endeavours, preferably living on the east coast of the United States, whose posts focused primarily on lifestyle content targeting a young adult audience. The authors independently identified six potential HI accounts. A pilot study was then conducted, asking 151 Instagram users to evaluate the similarity of each profile in terms of style, content, and expected follower characteristics to LM (44.5% female, average age 23.54, $SD = 4.23$). This second pilot group was also recruited via Prolific, but it was ensured that participants were different from those in the first pilot study outlined above. Participants were asked to indicate the extent to which they agreed (1 = strongly disagree, 7 = strongly agree) with the following statements: "These two influencers have a similar personal style", "These two influencers post similar content", and "These two

Table 1
Demographic characteristics of final sample

Characteristics	Number	Percentage of Sample[3]
Gender		
Male	130	61%
Female	80	37%
Don't identify as either male or female	4	2%
Age		
20-23	193	90%
24-26	17	8%
28 and over	4	2%
Ethnicities		
African	3	1%
Black	1	1%
Caribbean	5	2%
Western European	103	48%
East Asian	22	10%
Latino/Hispanic	6	3%
Mixed	6	3%
Native Indian	1	1%
South Asian	2	1%
North African	3	1%
White North American	2	1%
Mediterranean	22	10%
Eastern European	22	10%
Other	16	8%

[3] Some totals may not equal exactly 100% due to rounding effects

Table 2
Social media usage characteristics of final sample

Characteristics	Number	Percentage of Sample[4]
Frequency of Social Media Usage		
Never	1	1%
Monthly, but less than once per week	2	1%
Weekly, but less than daily	3	1%
Daily	45	21%
Several times a day	163	76%
Daily social media usage		
Less than 30 minutes	6	3%
30 minutes to 1 hour	37	17%
1-3 hours	114	53%
Over 3 hours	57	27%
Social Media Platforms used[5]		
Instagram	170	79%
Facebook	185	86%
LinkedIN	127	59%
Snapchat	115	54%
Twitter	77	36%
TikTok	57	27%
Whats App	114	53%
Other	32	15%

[4] Some totals may not equal exactly 100% due to rounding effects
[5] Participants were able to choose multiple responses



Fig. 1. Virtual Influencer, Lil' Miquela (LM)

influencers have similar followers". The authors then proceeded to review the Instagram profiles of the chosen VI and those HIs who were rated as 'most similar' on these criteria. The goal was to identify pictures of typical everyday situations that projected consistency across the VI and HI personas.

It was decided that the pictures posted by the HI who was rated as most similar in terms of style and followership (and second most similar in terms of content) were more sexually suggestive than the content posted by LM, making it difficult to identify pictures that were similar enough to avoid potential contamination effects. Consequently, this HI was eliminated from further consideration. We then proceeded to view the profile of the HI who was rated by the participants as second most similar in terms of style and followership, and first in terms of content. Upon closer comparative examination, it was decided that, given their predominant emphasis on fashion and beauty, this HIs pictures were similarly incompatible with the types of pictures posted on LM's profile. For all three criteria, the third most similar profile was that of Meghan DeAngelis (MD; Fig. 2). The authors reviewed her profile and determined that there were enough pictures of similar style and content to enable the creation of paired visual stimuli for LM and MD, in order to ensure the required consistency across the VI and HI personas. We then proceeded to identify specific images posted by both influencers in their accounts, with our initial screening resulting in 24 pairs of pictures. Upon further review and discussion, a final set of 6 photos were agreed upon to be used as experimental stimuli, given the similarities of the pictures in terms of context and comparable situations across the two influencers. Please see Appendix A for list of themes in the pictures and the set of photos displayed to participants.

4.2.2. Details of participation in focal experiment

Participation took place over two online sessions one week apart, in order to minimise the risk of common method bias (MacKenzie & Podsakoff, 2012). In session one, participants were asked to answer questions about their social media usage and demographic characteristics, and to complete the basic empathy scale (see below). A week after the first session, the link for the second session was made available to those who had successfully completed the first part of the study. At this stage, participants were randomly assigned to one of four experimental conditions ("human vs. virtual influencer" X "know vs. don't know the influencer's true nature"). Participants assigned to the "know" condition were told, "Below are several pictures of a social media influencer who is a real person (virtual personality) currently active on several social media platforms". In contrast, those in the "don't know" condition were told: "Below are several pictures of a social media influencer who is currently active on several social media platforms". Participants were then shown a collage of the six chosen photos of the respective influencer (Appendix A). After spending a minimum of one minute looking at the pictures, participants were asked to write a short description of each picture, which served as an attention check. As a manipulation check, participants in both conditions were asked whether the influencer they saw was a human or a virtual personality. In order to avoid any potential priming effects, this latter question only appeared at the very end of the survey for participants in the "don't know" condition.



Fig. 2. Human Influencer, Meghan DeAnglis (MD)

4.2.3. Measures

Unless otherwise indicated, all variables were measured on a 1 to 7 Strongly Disagree/Strongly Agree Likert scale.

Willingness to Follow. A single-item measure was employed to assess willingness to follow the influencer ("I would be willing to follow her using my personal social media account"), approximating a public representation of engagement with the influencer that is visible to others within one's social media network. We chose to use a single-item measure given that they have been found to be equivalent to multiple-item scales (Fisher et al., 2016) and the fact that we were measuring a specific behavioural intention, rather than a latent construct.

Attractiveness. In order to capture the importance of attractiveness in determining followership on social media (Djafarova & Rushworth, 2017) we also measured social attractiveness (5 items; sample item: "I think she could be a friend of mine"; $\alpha = .75$) and physical attractiveness (7 items; sample item: "I don't like the way she looks"; reverse coded; $\alpha = .85$) using scales adapted from McCroskey and McCain (1974).

Basic Empathy Scale. We used the basic empathy scale (BES), developed by Carré et al. (2013), which measures empathy along three dimensions: emotional contagion (6 items; sample item = "I get caught up in other people's feelings easily"; $\alpha = .73$); cognitive empathy (8 items; sample item = "When someone is feeling 'down' I can usually understand how they feel"; $\alpha = .68$[1]); and emotional disassociation (6 items; sample item = "My friends' emotions don't affect me much"; $\alpha = .80$).

Control Variables. Participants were asked to provide information about their age and gender (coded as Male = 1). We wanted to control for Instagram usage but were cognizant of the risk of priming participants to the purpose of the study. Therefore, instead of asking directly whether they were Instagram users, we asked them to choose the social media platforms they frequented regularly from a list of popular social media sites. This information was then coded as 1 for those reporting using Instagram regularly, and 0 otherwise.

[1] A closer inspection of these results showed that the removal of one item in the cognitive empathy subscale, "I can usually work out when my friends are scared", would increase the reliability to $\alpha = 0.74$. We re-ran all of the analysis with this attenuated cognitive empathy variable; however, this did not significantly change the results reported below. We therefore proceed to report the results using the full cognitive empathy scale.

5. Results

For simplicity of presentation, and given the potentially different cognitive processes involved, we split the data for analysis, seperately examining those participants who did not know the influencers' true nature, and those who did have this information. We used stepwise regression to test the relationships between our variables, as this allowed us to examine not only the overall interaction effects but also any incremental effects of adding control variables, independent variables, and the interaction terms. We entered age, gender, and the binary Instagram use variable as controls in the first step, followed by the independent variable(s) of interest and finally any interaction terms. All continuous variables were mean-centred prior to analysis (Aiken et al., 2003), and we used Dawson's (2014) recommendations for evaluating simple slopes when investigating visual graphs of interaction effects.

5.1. Manipulation & attention checks

In the condition where participants were aware of the influencer's true nature, a manipulation check was included after participants had viewed the influencer photos, asking whether the influencer was human or virtual. Any participant who answered this question incorrectly was not permitted to proceed any further to complete the experiment. To avoid the risk of participants guessing the purpose of the experiment, any participant who attempted the study again after having been removed had their responses deleted from the final data set. In the condition where participants were not informed about the true nature of the influencer, for those who were shown pictures of the human influencer, 77% correctly guessed that she was human; for those who were shown pictures of the virtual influencer, 46% guessed correctly that she was a virtual personality. In both conditions, participants were also asked to briefly describe the pictures they were shown. A visual inspection of these results did not identify any participants who raised suspicion of low attention during their participation.

5.2. Hypothesis testing in the "know" condition

Prior to formal hypothesis testing, we checked whether the influencer condition (human vs. virtual) had a direct effect on any of the dependent variables. As Table 3 shows, the experimental condition was only significant for social attractiveness, with the VI being more highly rated than the H1 ($\beta = 0.24$, $p < .05$).

To test H1, we interacted the human vs. virtual influencer condition with emotional contagion scores. As Table 4 shows, this interaction term was approaching significance for willingness to follow ($\beta = 0.25$, $p < .10$). Graphing these results in Fig. 3 shows that H1a is supported: participants scoring low on emotional contagion showed no preference for either influencer (simple slope = -0.11, n.s.), however those scoring high on emotional contagion reported a significantly higher willingness to follow the virtual influencer (simple slope = 1.18, $p < .05$). There was no significant effect of emotional contagion on perceptions of neither social nor physical attractiveness, leaving H1b unsupported.

To evaluate H2, Table 5 shows the coefficients on the influencer type X cognitive empathy interaction terms, which was significant for willingness to follow ($\beta = 0.37$, $p < .01$) and social attractiveness ($\beta = 0.31$, $p < .05$). The graphs for both of these results show similar patterns and are presented in Figs. 4 and 5. Participants low on cognitive empathy do not show a differential willingness to follow (slope = -0.47, n.s.) or perceptions of social attractiveness (simple slope = .06, n.s.) for either type of influencer. However, those who score high on cognitive empathy report significantly higher willingness to follow the VI (simple slope = 1.35, $p < .01$) and see her as more socially attractive (simple slope = 0.95, $p < .01$). Therefore, H2a and H2b were supported.

Table 6 shows the results of the regression analysis with influencer type interacted with emotional disassociation, in order to test H3. We see a significant effect of the interaction for willingness to follow ($\beta = -0.34$, $p < .01$) and social attractiveness ($\beta = -0.24$, $p = .05$), supporting H3a and H3b. Fig. 6 shows that, as predicted, participants who scored high on emotional disassociation showed no difference in willingness to follow either type of influencer (simple slope = -.43, n.s.). However, those scoring lower on this type of empathy are significantly more likely to follow the virtual influencer (simple slope = 1.57, $p < .01$). Fig. 7 shows similar results for social attractiveness: high emotional disassociation individuals showed no preference (simple slope = .09, n.s.), while those low on emotional disassociation found the VI significantly more socially attractive (simple slope: 0.92, $p < .01$).

5.3. Hypothesis testing in the "don't know" condition

In addition to the control variables included in the above analyses, we also included a binary variable to capture whether participants correctly guessed the influencers' true nature (see manipulation check section above). As in the know condition, we first looked at whether influencer type had an effect on any of the dependent variables, with no coefficients on the experimental condition showing significance in any of the regressions.

Additionally, the interaction of influencer type with each type of empathy only approached significance in the case of emotional disassociation and willingness to follow ($\beta = -0.27$, $p < .10$). However, graphing these results showed that neither of the slopes were significant. Therefore, H4 is supported in that none of the effects seen in the "know" condition were recreated in the "don't know" condition[2].

A summary of the hypotheses and results is shown in Table 7.

6. Discussion

6.1. Sweet escape

Our results show that individual differences in empathy may play an important role in engagement with human versus virtual agents in the context of social media. When participants are aware of the true nature of the agent, individuals scoring high on emotional contagion and cognitive empathy and low on emotional disassociation show a greater willingness to follow the VI. Additionally, those scoring high on cognitive empathy and low on emotional disassociation believed the VI would be a better fit with their social circle (social attractiveness). Overall, these findings are consistent with an "escapism effect", where the VI is expected to provide greater diversionary benefits from everyday human emotional experiences and require fewer cognitive resources in the form of emotional sense making. This effect manifests through a preference for VIs, with their authentically fake storylines, over HIs, with authentic, human ones.

In the case of cognitive empathy and emotional disassociation, the same patterns were found for social attractiveness; these are the two dimensions of empathy with the greatest long-term consequence, as they may affect cognitive resources available for other activities and long-term emotional experiences, respectively. The construct of social attractiveness, as measured here, captures the willingness to interact with the focal person and/or include them in one's social circle; these results may indicate an even stronger escapism effect, in that the expected lower emotional entanglements of the VI may even gain them (hypothetical) access to users' social circles. Social identity theory postulates that individuals will want to be affiliated with others who bring benefits to the individual's social standing (Jin, 2018; Tajfel & Turner, 1986). Although such social identity effects were outside the scope of this study, it is possible that these inclusion criteria may also include benefits to the individual that extend beyond their external social status – e.g., lower empathetic responses.

[2] In the interest of brevity, detailed results for this analysis are not presented here but are available from the corresponding author upon request.

Table 3
Regression results of effect of influencer type when participants know influencer is human or virtual ("know" condition)

	DV: Willingness to Follow			DV: Social Attractiveness			DV: Physical Attractiveness		
Variable	B	SE B	β	B	SE B	β	B	SE B	β
Age	-.09	.13	-.07	-.01	.07	-.02	.02	.07	.02
Gender	-.52	.38	-.14	-.33	.22	-.14	-.58	.21	-.27**
IG Use	.01	.52	.00	-.12	.30	-.04	.21	.28	.07
Influencer	.56	.36	.15	.51	.21	.24*	.36	.20	.18

n = 105
Gender: 0 = Female, 1 = Male
IG Use: 0 = Do not use Instagram, 1 = Use Instagram regularly
Influencer: 0 = Human Influencer, 1 = Virtual Influencer
* p < .05,
** p < .01

Table 4
Regression results showing effect of interaction of influencer type and emotional contagion when participants know influencer is human or virtual ("know" condition)

	DV: Willingness to Follow			DV: Social Attractiveness			DV: Physical Attractiveness		
Variable	B	SE B	β	B	SE B	β	B	SE B	β
Fixed Effects									
Age	-.08	.13	-.07	-.01	.07	-.01	.02	.07	.03
Gender	-.51	.45	-.13	-.42	.27	-.18	-.61	.25	-.28*
IG Use	.06	.51	.01	-.09	.31	-.03	.21	.29	.07
Influencer	.54	.36	.15	.49	.21	.23*	.35	.20	.17+
Emotional Contagion	-.43	.26	-.23	-.18	.16	-.16	-.04	.15	-.04
Interaction Effects									
Influencer X Emotional Contagion	.66	.37	.25+	.13	0.22	0.08	.02	.21	.01
Overall F	1.51			1.61			2.14		
Adjusted R^2	.03			.03			.12		
Δ F	3.22+			.37			.01		
Δ R^2	.03			.00			.00		

n = 105
Gender: 0 = Female, 1 = Male
IG Use: 0 = Do not use Instagram, 1 = Use Instagram regularly
Influencer: 0 = Human Influencer, 1 = Virtual Influencer
+ p < .10,
* p < .05,
** p < .01

There was no effect of any of the three empathy dimensions on physical attractiveness ratings. This is not altogether surprising, as physical attractiveness is a cognitive evaluation of the target's physical appearance, which may not have a clear link to behavioural intentions (Santiago & Serralha, 2022), nor intention to interact with the influencer in the future (Sokolova & Kefi, 2020). Therefore, in the context of empathetic reactions, there is no real emotional consequence to judgements of physical attractiveness. Additionally, the limited exposure to the influencers in restricted contexts, and using pictures that were clearly chosen (by the influencer) to be attractive may have limited the variability in physical attractiveness ratings, thereby precluding any differences across conditions or sub-groups.

6.2. Disclosing virtuality

As expected, all of these relationships disappeared when participants were not aware of the influencer's nature. This again supports the existence of the escapism effect, that when participants are aware that the influencer may offer a less realistic emotional experience, they show a preference for engaging with that influencer. Importantly, however, simple suspicion about the influencer's nature does not appear to be enough; the VI presented in the experiment is ambiguous enough to raise doubt regarding her true nature (Arsenyan & Mirowska, 2021). Yet, such doubt was not enough to engage empathy to alter reactions. Only when participants were explicitly aware of her virtual nature were they able to indulge in her profile with confidence that a less intense emotional experience, and therefore an escape from the emotional entanglements of their daily lives, awaited them.

These results also provide further support to earlier work that pitted the Computer Are Social Actors Paradigm (CASA; Nass & Moon, 2000) against the UVH in the context of HIs vs VIs (Arsenyan & Mirowska, 2021). Although under CASA we would expect VIs to be treated similarly to their human counterparts, as in this prior research, we see that this is not the case here. Given the lack of consensus in recent CASA studies studying digital entities on social media (Meng & Dai, 2021; Park et al., 2021; Spence et al., 2019), the specific context of social media and user pursuit of diversion in this realm may act as a boundary condition to the treatment of digital agents as humans (Nowak et al., 2015).

The current work also sheds a more favourable light on some potential consequences of the UVH. Although the virtual agent may elicit some feelings of creepiness and negative emotions (Arsenyan & Mirowska, 2021; Mori, 1970), there may be an upside to these reactions in the form of greater diversionary benefits. The virtual agent with a high human resemblance, but who maintains their virtual status, may provide just enough reality to allow for users to be transported into the virtual agents' world (Brown, 2015), but also allow the user to dehumanise or distance themselves just enough to protect themselves from the emotional repercussions of positive, affiliative reactions.

Worral et al. (2021) have called for the recognition of a social-emotional paradigm in studying user motivations to engage in online communities. We echo this call and add to it a consideration of the coherence of existing norms and practices with the various profiles of potential other users – human vs. virtual, and in the latter case the degree of autonomy – in studying engagement, information-sharing, and social and emotional support in online social contexts. Much has been written of the potential negative effects of social media exposure,

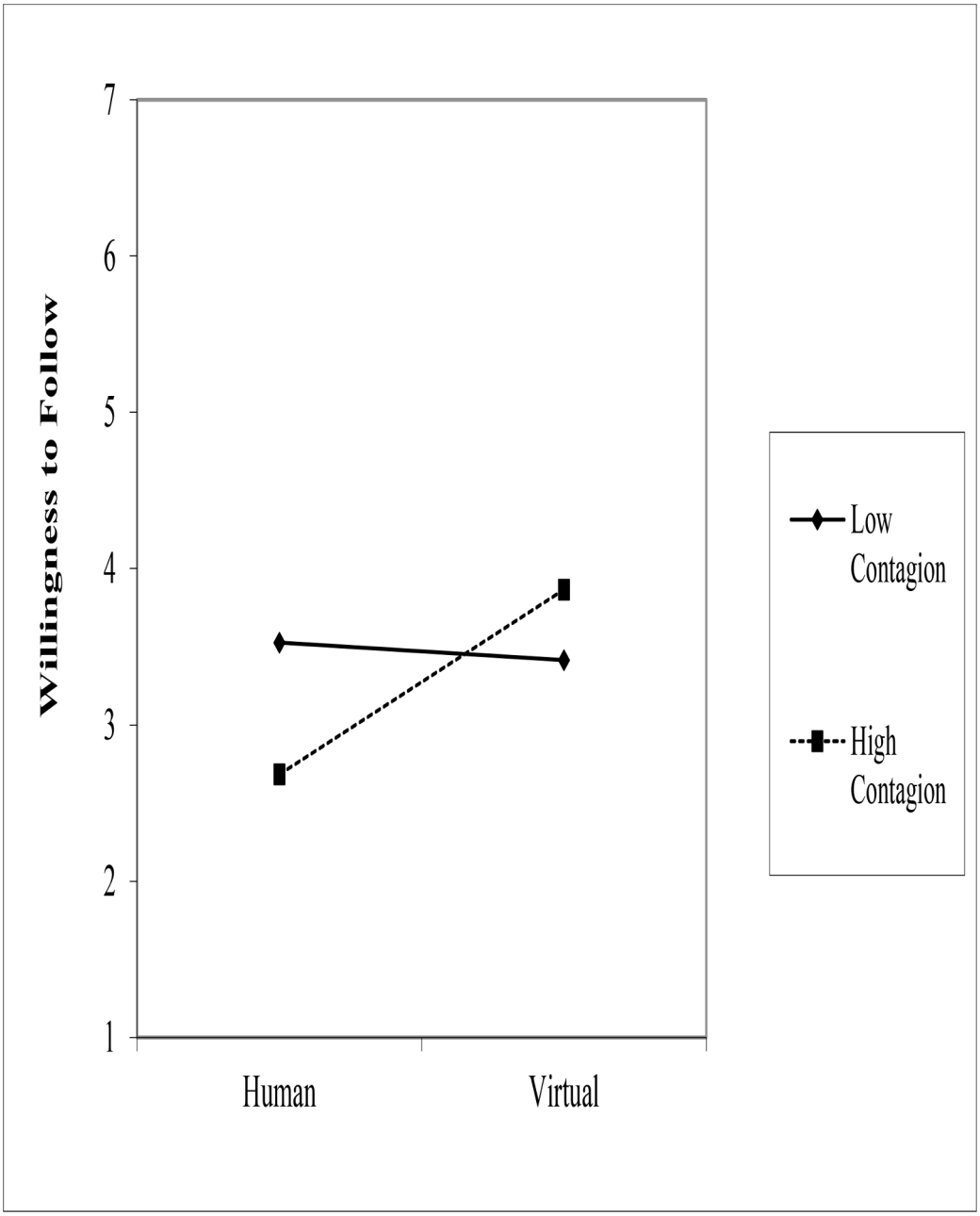


Fig. 3. Interaction of influencer type with emotional contagion on willingness to follow influencer when know influencer is human or virtual

particularly on young adults and their emotional and psychological well-being (Boer et al., 2022; Schemer et al., 2021; Schmuck, 2021; Valkenburg et al., 2022). However, our results imply that the presence of virtual agents in human social networks may require a reconsideration of the exact role of individual differences (Nowak et al., 2015) such as self-esteem (Djafarova & Rushworth, 2017; Schmuck, 2021), empathy, or shyness (Birnie & Horvath, 2002; von der Pütten et al., 2010) in the study of social media usage and consequences. Between human users, empathy serves to build connections (Vossen & Valkenburg, 2016) and parasocial relationships (PSR; Hwang & Zhang, 2018) by allowing users to relate to each other; however, between human and virtual users, it seems that empathy builds connections not via relatedness, but via the escape potential of interacting with a non-human agent. Thus, empathy is now seen as not necessarily an asset but a potential liability, while virtual agents become not just a robotic presence but also a possible safe haven that has the potential to buffer vulnerable users from some of the detrimental influence of social media influencers (Vossen & Valkenburg, 2016).

6.3. *Practical implications*

These results have practical implications for individuals and companies looking to leverage the power of influencers, human or virtual, to connect with their target audience via social media. For individuals who are not yet followers of a given VI, trying to play on empathy may backfire. VIs may be expected to have less rich emotional lives, or to elicit less genuine empathetic reactions, which may explain part of their appeal. Trait activation theory (Tett & Guterman, 2000) argues that certain dispositional traits are only activated when necessary given the context. Highly emotional content, although intended to build relationships with the audience, may actually turn off potential followers by overly activating empathetic responses; the presence of the escapism effect may mean that users want a break from empathising with others, and the associated increased cognitive load (Cameron et al., 2019).

Table 5
Regression results showing effect of interaction of influencer type and cognitive empathy when participants know influencer is human or virtual ("know" condition)

	DV: Willingness to Follow			DV: Social Attractiveness			DV: Physical Attractiveness		
Variable	B	SE B	β	B	SE B	β	B	SE B	β
Fixed Effects									
Age	-.03	.12	-.02	-.02	.08	-.03	-.01	.07	-.01
Gender	-.65	.36	-.17^{+}	-.30	.23	-.13	-.51	.21	-.24
IG Use	.22	.49	.04	-.13	.31	-.04	.14	.29	.05
Influencer	.44	.33	.12	.51	.21	.24*	.39	.20	.19
Cognitive Empathy	-1.38	.33	-.58**	-.39	.20	-.27^{+}	-.09	.19	-.07
Interaction Effects									
Influencer X Cognitive Empathy	1.20	0.44	.37**	.59	.28	.31*	.36	.26	.20
Overall F	3.97**			2.25*			2.58*		
Adjusted R^2	.15			.07			.08		
Δ F	7.39**			4.52*			1.90		
Δ R^2	.06			.04			.02		

n = 105
Gender: 0 = Female, 1 = Male
IG Use: 0 = Do not use Instagram, 1 = Use Instagram regularly
Influencer: 0 = Human Influencer, 1 = Virtual Influencer
$^{+}$ $p < .10$,
* $p < .05$,
** $p < .01$

Another implication is that, to leverage this escapism effect of VIs, their true nature should be clearly advertised. Although this may appear to be a counterintuitive recommendation, VIs that "pass" for human may come with the same expected emotional baggage as HIs. High-empathy potential followers may appreciate the assurance of a less authentic emotional experience as a way to minimise their emotional investment in the influencer and their storylines. This recommendation also builds on Arsenyan and Mirowska's (2021) findings with respect to the UVH, where the unambiguously non-human influencer received positive reactions comparable to the HI, but the more ambiguous, human-like VI received much more negative reactions.

A further implication is that, if individuals are seeking an escapism effect from social media, messages concerning emotionally charged content should be carefully calibrated and transmitted. Highly emotionally charged content, particularly from human influencers, may lead to "emotional overload" where attentional processes are disrupted and the intended message may be filtered out as a protective mechanism (Ytre-Arne & Moe, 2021). If highly empathetic individuals avoid such content or profiles that they believe tend to display such content, then such messaging may need to be transmitted via different channels or methods.

Conversely, it appears that ratings of physical attractiveness are immune to the effect of individual level empathy. Therefore, highly emotional content that is meant to elicit positive visual evaluations only may be appropriate. However, the stimuli chosen to convey such messaging (e.g., phonological, lexical, grammatical, denotational; Cardona-Rivera et al., 2020) should be carefully considered to avoid any unintended consequences.

Finally, Senft (2013) argues that becoming a celebrity requires the transformation from "citizen to corporation" (p. 351). For the human influencer, this happens in the form of self-branding, where the product being proposed is intimacy with the influencer, and the price being asked is follower attention. For this exchange to be successful, followers need to be emotionally invested in the influencers' life and narratives. Although questions have been raised about the potential economic, legal, and social consequences of such exchanges (Abidin, 2018), the presence of humans on both sides of the transaction makes things rather straight-forward to understand. VI's on the other hand, are commodifying intimacy with, by their very nature, fake storylines and fake people. As the results of the current research show, there may be sub-groups of social media users who will be more drawn to this "authentically fake" storytelling, which opens up additional questions of potential real-world consequences arising from virtual-world events.

For instance, reactions to highly emotional content for certain categories of individuals may have far-reaching consequences in terms of the types of narratives that are shared and valorised in the wider social media realm, particularly as online activists rely on users' ability to empathise with their cause (Martínez García, 2020). Senft (2013) argues that online identity is no longer the property of the individual, but now resides with the "perceiver", taking the power of dissemination away from the storyteller and granting it to the receiver of this narrative, in line with the received narrative argument from Castricato et al. (2021). As algorithms become "key factors in how attention is distributed" (Marwick, 2018, p. 163), the individuals and narratives that garner more attention online will find themselves in a virtuous cycle of follower growth, leading to greater market value and income-generating opportunities for the influencer (Abidin, 2018). There are two potential consequences of this process that we believe deserve highlighting. On the one hand, highly emotionally charged content recounting difficult or unjust circumstances may be ignored in favour of more frivolous fare, particularly as algorithms adapt to preferences among highly empathetic individuals. However, if this is also the same sub-population more likely to engage in offline action to counter injustices or difficulties (Craddock, 2020), then this activism may end up muted in the long run. Alternatively, if virtual influencers have an advantage in garnering attention for highly emotional content amongst this same sub-group of social media users, then the causes that become visible on social media (and, potentially, acted upon offline) may be decided by the corporations (and potentially their algorithms) behind these virtual personas rather than by the human users of the platform. Future research should draw on theories and techniques within anthropology, sociology, race and gender studies, media and cultures studies (Senft, 2013) to consider such potential long-term and offline effects.

6.4. Limitations

The findings of this study are limited to the initial perception of the virtual agents, as we purposely did not sample followers of the focal influencers in our study. Although this was by design, it means that our study does not capture the reduced uncertainty and increased trust required in the human-agent interaction (B. Liu, 2021) that may develop as a result of sustained interactions (Bickmore & Picard, 2005). Therefore, the present study is not able to study the differences in longer-term outcomes of user-HI vs. user-VI interactions, such as PSR formation. Given that empathy is an interpersonal skill that helps to build relationships (Cameron et al., 2019), including parasocial ones

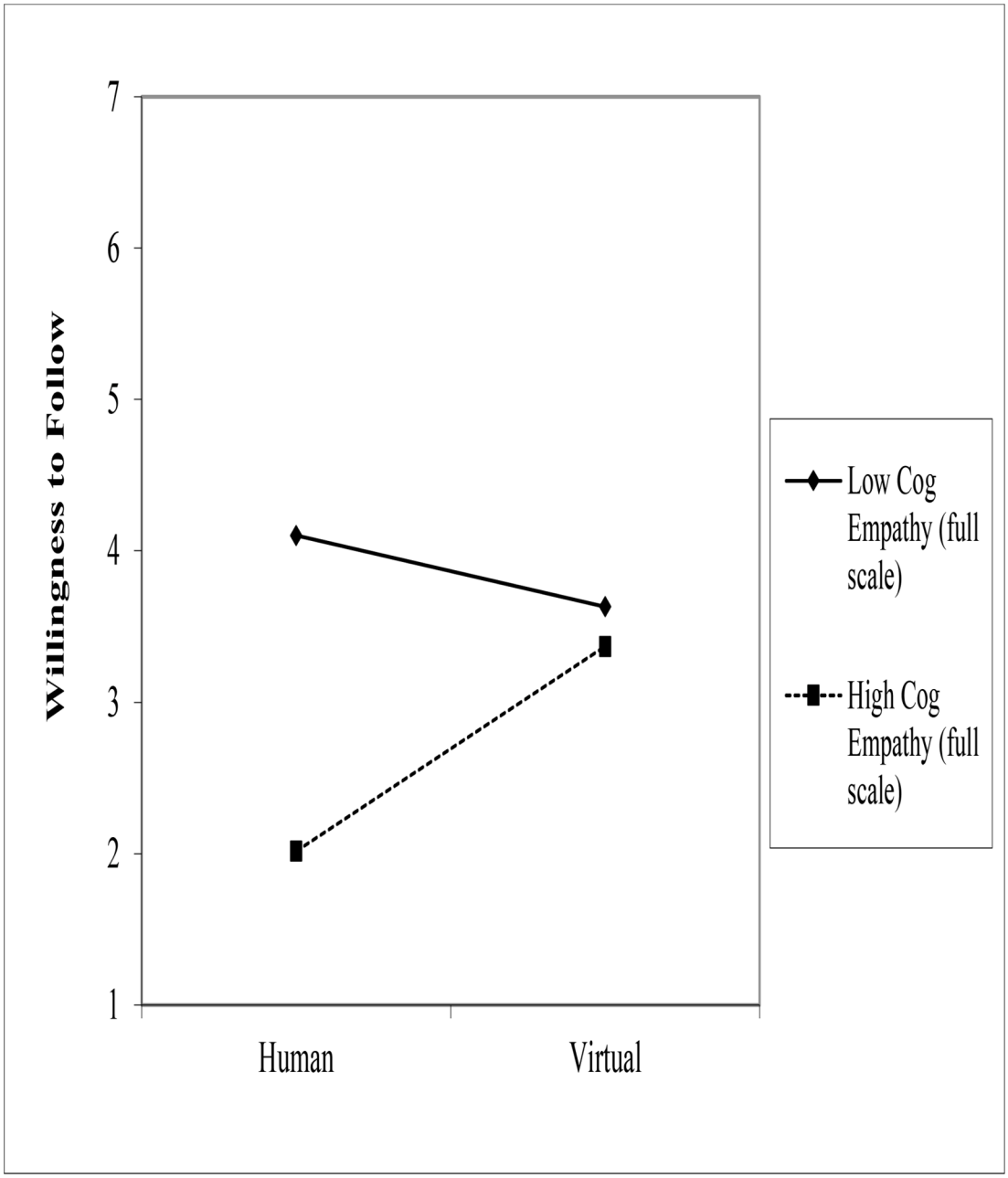


Fig. 4. Interaction of influencer type with cognitive empathy on willingness to follow influencer when know influencer is human or virtual

(Hartmann, 2017; Hwang & Zhang, 2018), it is possible that the propensity for higher empathy participants to follow VIs indicates a potential budding PSR. However, PSR requires parasocial interactions over time, which we were not able to capture in the current study.

Additionally, the limited experimental stimuli of one human and one virtual influencer and the exclusion of participants who were familiar with the focal influencers means we are not able to consider any effects of persona popularity or social status markers on willingness to interact with a virtual influencer on a publicly visible social media platform. Interestingly, social attractiveness was the only dependent variable significantly affected by the experimental manipulation, with the virtual influencer being seen as more socially attractive. Future longitudinal studies should consider how social identity related characteristics such as social standing, popularity, or social image and their expected benefits may affect engagement with human versus virtual influencers.

Another limitation is the use of pictures in the experiment, as opposed to videos or written content. This was by design to minimise potential confounding effects of text interpretation or video playback issues, and to create a stimulus very close to the majority of content on Instagram. However, as non-verbal behaviour and attitudes have been found to elicit positive impressions in virtual human-agent encounters (Cafaro et al., 2016), a replication of this study with videos would allow for an exploration of the potential moderating effect of the first encounter medium.

Our data collection relied on a young adult, student population, who participated in an online survey. Although this sample was representative of the demographics of Instagram users, we acknowledge that the implications of this study may not be appropriately extrapolated to other user populations or social media platforms. Furthermore, researchers' choices pertaining to which virtual and human influencers and stimuli were included in the pilot study and final experimental materials may reflect unconscious biases arising from our specific backgrounds and experiences (Das et al., 2022). For instance, deciding to exclude one of the "more similar" human influencers due to what we perceived to be

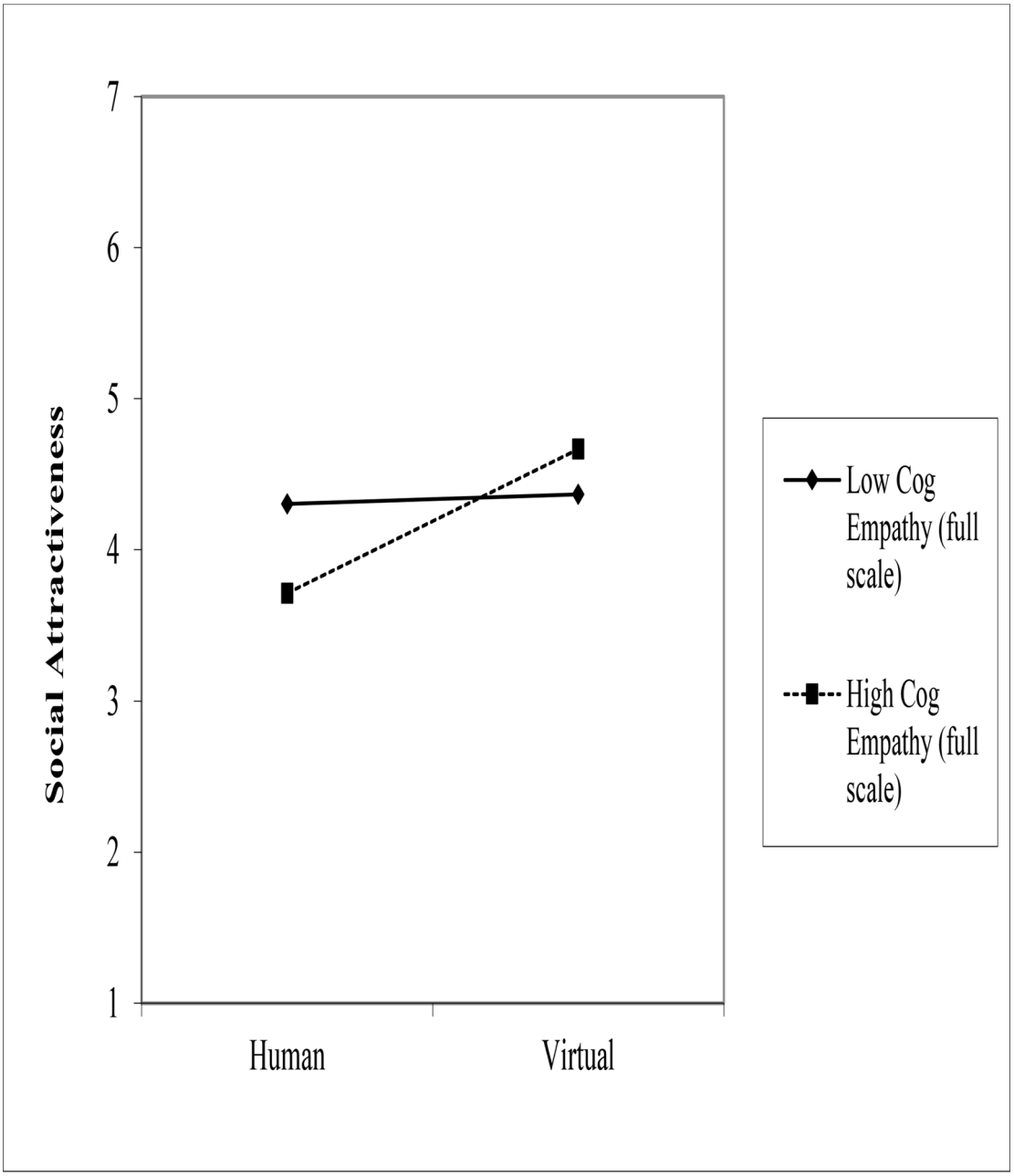


Fig. 5. Interaction of influencer type with cognitive empathy on social attractiveness to follow influencer when know influencer is human or virtual

overly sexualized content means that the stimuli ultimately chosen for the final experiment are moderately explicit at most. Although we acknowledge that it is not possible to eliminate all such biases, all attempts were made to develop the materials, collect, analyse, and report data as objectively as possible. The difficulty that we found in matching up the highly sexualized human profile with the virtual one may also indicate a potential boundary condition for virtual influencers' mimicking of human social media content, as well as interesting potential cross-cultural differences. For example, developers may be simultaneously relying on common social taboos against sexual intimacy with inanimate objects (Su et al., 2019) and the acceptability of emotional attachment to inanimate objects (Sharabiani, 2021) to allow their VIs to gain popularity and followers. However, if similar norms do not hold in other cultural contexts (Motschenbacher, 2018), then this approach may perpetuate the Anglocentrism of studies of internet-based phenomenon (Marwick, 2018).

Additionally, although several steps were taken to minimise the risk of common method variance (CMV) as a driver of the results, it is still possible that some CMV effects persisted if participants recalled their participation from one part of the study to another. However, the finding of several significant moderation results reduces CMV concerns, given the difficulty of identifying moderation results in its presence (Siemsen et al., 2010).

Finally, one general limitation in current VI research is the limited knowledge that the public and academics have regarding the mechanisms that operate behind the virtual influencers - i.e. whether their content is manually authored, automatically generated, or some hybrid methodology. The current results, therefore, rely on individuals' personal beliefs about how a "virtual influencer" operates, which may interact with their own ideas, beliefs, and fears regarding artificial intelligence and automated systems.

Table 6
Regression results showing effect of interaction of influencer type and emotional disassociation when participants know influencer is human or virtual ("know" condition)

	DV: Willingness to Follow			DV: Social Attractiveness			DV: Physical Attractiveness		
Variable	B	SE B	β	B	SE B	β	B	SE B	β
Fixed Effects									
Age	-.10	.12	-.08	-.03	.07	-.04	-.01	.07	-.02
Gender	-.53	.37	-.14	-.27	.23	-.12	-.46	.21	-.21
IG Use	.09	.50	.02	-.09	.30	-.03	.23	.28	.08
Influencer	.57	.34	.16	.51	.21	.24*	.34	.19	.16
Emotional Disassociation	.62	.21	.36**	.12	.13	.12	-.09	.12	-.10
Interaction Effects									
Influencer X Emotional Disassociation	-.94	.33	-.34**	-.39	.20	-.24*	-.28	.19	-.18
Overall F	2.74*			2.09+			3.41**		
Adjusted R^2	.09			.06			.12		
Δ F	8.10**			3.86*			2.24		
Δ R^2	.07			.04			.02		

n = 105
Gender: 0 = Female, 1 = Male
IG Use: 0 = Do not use Instagram, 1 = Use Instagram regularly
Influencer: 0 = Human Influencer, 1 = Virtual Influencer
+ p < .10,
* p < .05,
** p < .01

7. Conclusion

This study contributes to research into interactions with virtual agents by considering the role of empathy as an individual difference variable predicting preferences for human vs. virtual influencers in a social media context. Results show that there is no difference in the initial perception of the agent with regards to their human counterpart when the virtual nature of the agent is not disclosed to the user. When the virtual agent is clearly described as such, we observe an escapism effect, where users turn to virtual influencers as a sweet escape from the potentially excessive emotional experience of interacting with fellow humans. These findings have implications for the development, deployment, and employment of virtual influencers, as well as the understanding of proliferation of highly emotionally charged content in social media settings. We believe that this research provides insight for future virtual influencer, as well as virtual agent research, making a case for the importance of empathy in engaging with these digital entities.

Future research should investigate the emotional experiences of individuals, both when choosing which influencers to follow and when interacting with their chosen influencers on social media. Diary studies or interviews could be utilised to explore these interactions in more depth. Additionally, the potential role of virtual agents in offsetting some of the established negative effects of social media usage, such as low self-esteem, excessive social comparisons, or depression (Alfasi, 2019), represents a fertile area of study. As AI technologies and, by extension, virtual agents, continue to evolve, insights into automated story generation (e.g., Riedl, 2010) may allow for new methods of exploring human engagement with the interactive storylines generated by virtual influencers.

Finally, if the escapism effects embodied by a willingness to follow a VI represents a defensive psychological mechanism (Labrecque et al., 2011), then future research should explore whether highly empathetic followers of VIs are more likely to dehumanise them than low empathetic followers, implying that such defence mechanisms may be deployed collectively for greater psychological protection.

CRediT authorship contribution statement

Agata Mirowska: Conceptualization, Methodology, Validation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition. **Jbid Arsenyan:** Conceptualization, Methodology, Validation, Data curation, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Visual Stimuli Presented to Participants

Table
Virtual Influencer:

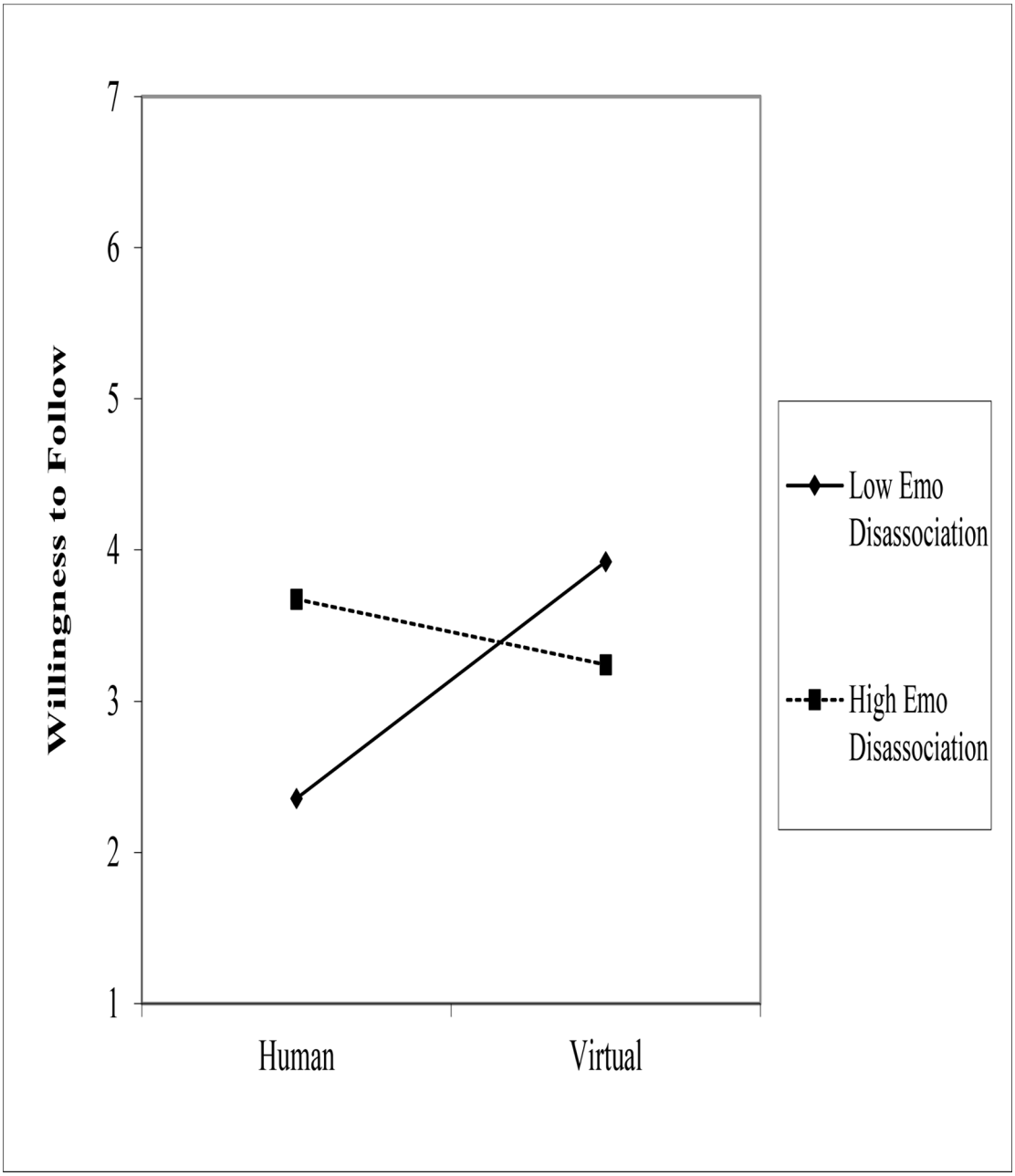


Fig. 6. Interaction of influencer type with emotional disassociation on willingness to follow influencer when know influencer is human or virtual

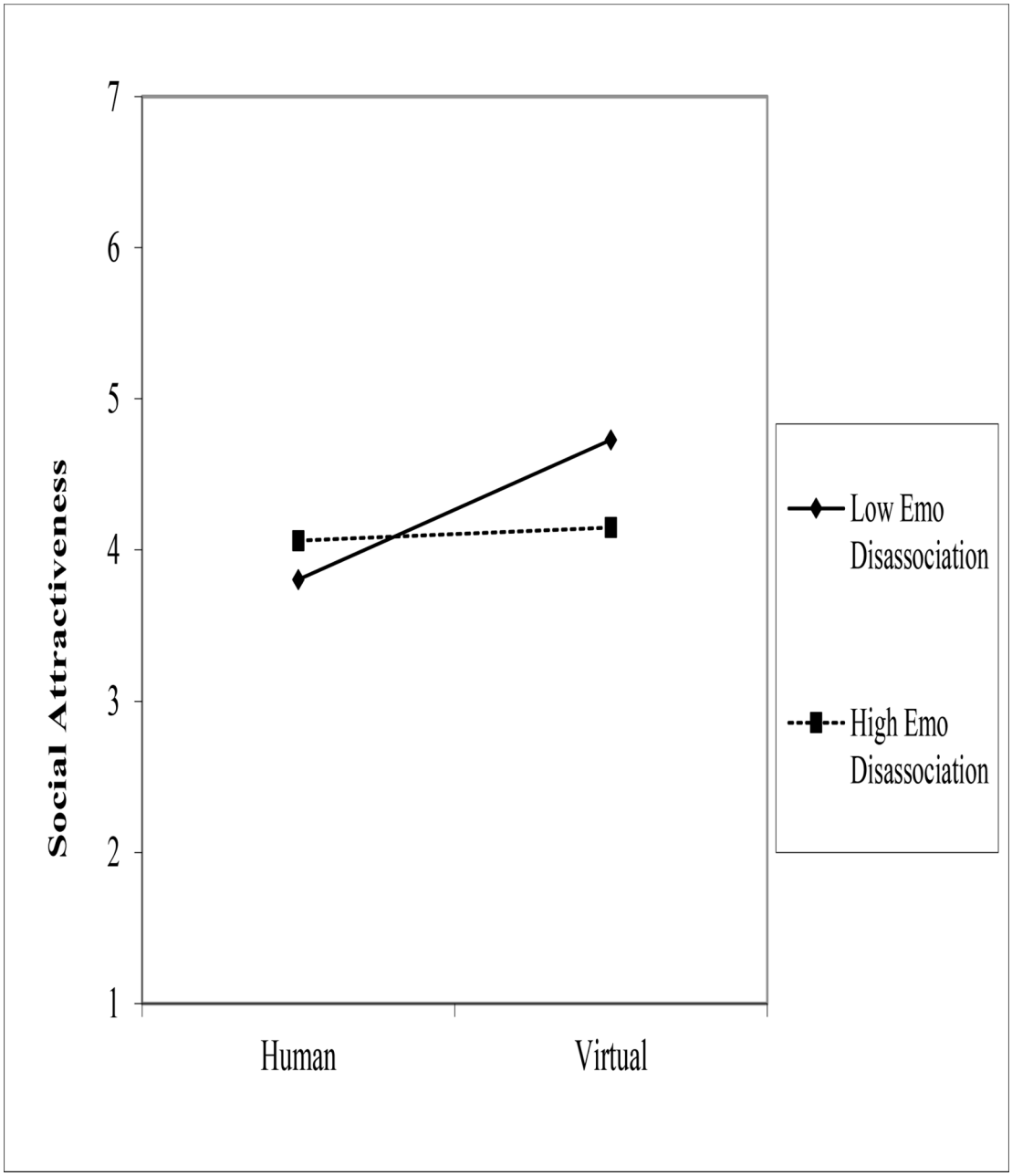


Fig. 7. Interaction of influencer type with emotional disassociation on social attractiveness to follow influencer when know influencer is human or virtual

Table 7
Summary of Hypotheses

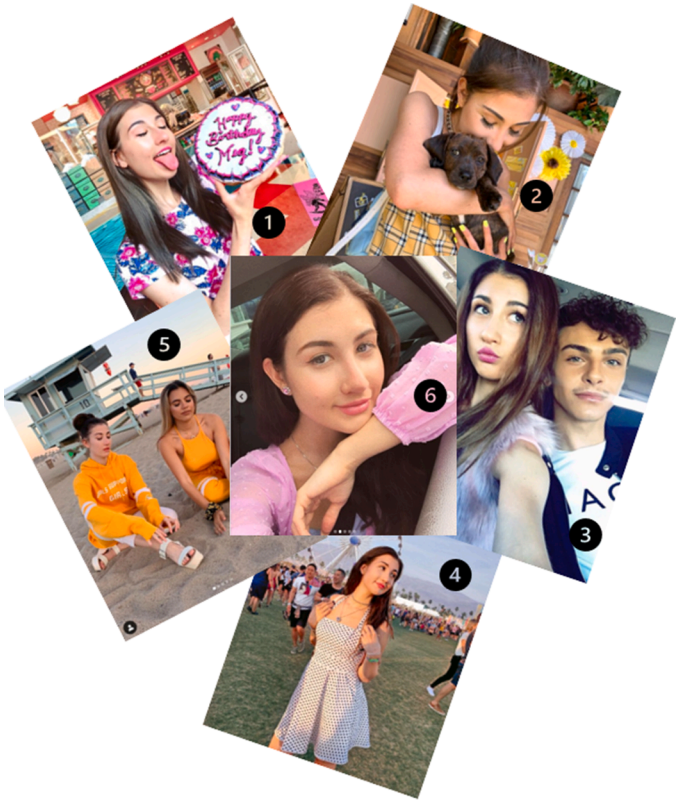
No.	Relationship Tested	Result	Summary
H1a	Emotional Contagion moderates Influencer Type → Willingness to Follow	Supported	Individuals high on emotional contagion report a higher willingness to follow the virtual, rather than human influencer; individuals low on emotional contagion report similar willingness to follow the human versus virtual influencer.
H1b	Emotional Contagion moderates Influencer Type → Social Attractiveness	Unsupported	No difference in social attractiveness ratings for human versus virtual influencer based on participant emotional contagion scores
H1b	Emotional Contagion moderates Influencer Type → Physical Attractiveness	Unsupported	No difference in physical attractiveness ratings for human versus virtual influencer based on participant emotional contagion scores
H2a	Cognitive Empathy moderates Influencer Type → Willingness to Follow	Supported	Individuals high on cognitive empathy report a higher willingness to follow the virtual, rather than human influencer; individuals low on cognitive empathy report similar willingness to follow the human versus virtual influencer.
H2b	Cognitive Empathy moderates Influencer Type → Social Attractiveness	Supported	Individuals high on cognitive empathy rate the virtual influencer as being more socially attractive, as compared to the human influencer; individuals low on cognitive empathy rate the human and virtual influencers similarly in terms of social attractiveness
H2b	Cognitive Empathy moderates Influencer Type → Physical Attractiveness	Unsupported	No difference in physical attractiveness ratings for human versus virtual influencer based on participant cognitive empathy scores
H3a	Emotional Disassociation moderates Influencer Type → Willingness to Follow	Supported	Individuals low on emotional disassociation report a higher willingness to follow the virtual, rather than human influencer; individuals high on emotional disassociation report similar willingness to follow the human versus virtual influencer.
H3b	Emotional Disassociation moderates Influencer Type → Social Attractiveness	Supported	Individuals low on emotional disassociation rate the virtual influencer as being more socially attractive, as compared to the human influencer; individuals high on emotional disassociation rate the human and virtual influencers similarly in terms of social attractiveness
H3b	Emotional Disassociation moderates Influencer Type → Physical Attractiveness	Unsupported	No difference in physical attractiveness ratings for human versus virtual influencer based on participant emotional disassociation scores
H4	H1, H2, H3 only supported in "know" condition	Supported	The moderating effects of emotional contagion, cognitive empathy, and emotional disassociation were only supported when participants were expressly told that the influencer was human or virtual

Table 8
Picture Descriptions

1	Influencer's birthday
2	Influencer with a dog
3	Influencer with a male friend
4	Influencer at an open-air festival
5	Influencer at the beach with a female friend
6	Influencer's portrait



Human Influencer:



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